TAYLKOMB modular comb system: engineering validation study

The TAYLKOMB system enters a confirmed white space — no modular comb system exists on the market — but its current assembled length of 405–424 mm is roughly twice the length of any professional comb benchmarked, and its PA66-GF30 handle material poses a serious dimensional stability risk in wet salon environments. These two issues demand redesign before CAD finalization. The system's weight target (22–35 g), connector geometry, POM comb-body material choice, and 1.3 mm rat-tail tip are all validated by competitive benchmarks and manufacturing data. Below is the full engineering analysis across six research domains, with point-by-point validation verdicts at the end.

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1. Competitive benchmarking reveals a tight dimensional envelope

Professional cutting combs cluster in a remarkably narrow band. Y.S. Park's cutting-comb line (models 332, 334, 337, 339, 402, 452) ysparkusa spans **181–202 mm** in overall length and weighs approximately **9–10 g** per comb body. YS Park +4 Their tail combs (101, 102, 106, 111, 132) run **216–245 mm** overall — the longest being the 106 Extra Long Tail at 245 mm. Japan Pro Tools All Y.S. Park combs use **PEI/Ultem polyetherimide** with a stated heat resistance of **220 °C (428 °F)** YS Park +3 and feature Gradual Decreasing Pitch (GDP) teeth that change spacing by 10 µm per tooth. Japan Pro Tools +2

Sam Villa Signature Series combs use a **carbon-fiber/silicone/graphite resin composite** rated to **232 °C (450 °F)**. SalonCentric Minerva Beauty The Short Cutting Comb measures **200 mm (7.875 in)** Sam Villa +2 and weighs **17 g**; the Long Cutting Comb is **222 mm (8.75 in)** Sam Villa SalonCentric at **20 g**. Hercules Sägemann uses **vulcanized ebonite (hard rubber)** for its premium line Amazon Hercules-saegemann and **POM and polycarbonate** for injection-molded models, Amazon with weights of **20–40 g**. Krest Goldilocks combs are notably made from **DuPont Delrin acetal copolymer** — the exact POM family proposed for TAYLKOMB comb bodies Pearlon — validating POM as a proven professional-comb resin.

Brand/Model	Length (mm)	Weight (g)	Material	Heat Rating	Price (USD)
				220	$10–

Y.S. Park 339	181	~9	PEI/Ultem	°C	20
Y.S. Park 332	186	~9	PEI/Ultem	220 °C	$13–26
Y.S. Park 337	190	~9	PEI/Ultem	220 °C	$12–24
Y.S. Park 452	202	~10	PEI/Ultem	220 °C	$14
Y.S. Park 101 (tail)	216	—	PEI/Ultem	220 °C	$5–11
Y.S. Park 106 (tail)	245	—	PEI/Ultem	220 °C	$7–14
Sam Villa Short Cutting	200	17	Carbon/silicone/graphite	232 °C	$7.50
Sam Villa Long Cutting	222	20	Carbon/silicone/graphite	232 °C	$7.50
Sam Villa Wide Cutting	216	~20	Carbon/silicone/graphite	232 °C	$7.50
Hercules IO 7 Handle	203	30–40	Ionized thermoplastic	—	$10
Krest G1 Sectioning	210	—	Delrin (POM)	—	$3–8
Kent A 16T	185	—	Cellulose acetate	—	$12–18
Mason Pearson C2	197	~45	Cellulose acetate	—	$38
Fromm F3023	216	~18	Silicone/ceramic	177 °C	$6–9

The competitive envelope for professional cutting combs is **181–222 mm body length** and **9–30 g weight**. TAYLKOMB's Main Comb body at **208 mm** sits comfortably within this band. Its comb-body-only weight will be well within the competitive range if manufactured in POM.

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2. No modular comb system exists — validated white space

Extensive searching confirms **no modular interchangeable-head comb system is on the market as of 2026**. The closest analogues found were a hair-dye-dispensing device with swappable comb heads Google Patents (US Patent 8,118,036, using an elastic polymer nipple connector — not a styling tool), clipper-blade interchange mechanisms from the 1950s (US Patent 2,860,412), Google Patents and powered styling-tool attachments (Dyson Airwrap, Shark SilkiPro). None of these offer a manual modular comb for professional styling.

Cross-industry modular systems reveal proven connector strategies directly applicable to TAYLKOMB. The **Dyson Airwrap** uses a spring-loaded latch plus twist-lock for wobble-free attachment swaps, mistywest with backward compatibility across all product generations. **Safety razors** converged on an M5 × 0.8 mm threaded stud as a de facto standard Damn Fine Shave — demonstrating that an industry can standardize on one interchangeable interface. **Dental instruments** (Smile Line modular system, Dentanomic) use screw-in or press-fit connections with O-ring seals, designed for thousands of sterilization cycles. **Precision screwdrivers** (iFixit, Wiha ClicFix, PB Swiss Click) use magnetic sockets, mechanical click-locks, and spring-steel hex interfaces — all rated for lifetime use. PB Swiss Tools PB Swiss Tools

The most relevant connector mechanisms for TAYLKOMB rank as follows:

- **Ball-detent (recommended primary)**: Fast engagement (~1 second), moderate holding force, proven for 10,000+ cycles with steel ball on POM detent
- **Twist-lock**: Excellent wobble resistance, proven by Dyson, but adds rotational complexity
- **Click-lock**: Positive tactile/audible engagement, proven by Wiha and PB Swiss at lifetime-rated cycles
- **Magnetic**: Infinite cycle life but poor torque resistance without a secondary mechanical interlock

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3. Material selection needs one critical change

POM for comb bodies is validated. POM copolymer offers **fatigue endurance of 23 MPa at 10^7 cycles**, tensile strength of 61 MPa, Midlandplastics excellent dimensional stability (water absorption only **0.22%**), Midlandplastics and proven thin-wall moldability down to **0.75 mm**. Krest Goldilocks and Hercules Sägemann both use POM for production combs, confirming commercial viability. Pearlon Moser POM's Barbicide resistance is rated

excellent — it withstands repeated immersion in dilute quaternary ammonium solutions without degradation. Its continuous service temperature of **82–100 °C** is adequate for combs that experience brief hot-tool contact rather than sustained heat. MakerStage

PA66-GF30 for handles is problematic. While PA66-GF30 offers superior stiffness Nylon Plastic (flexural modulus **6,500 MPa** lookpolymers vs. POM's **2,585 MPa**) and excellent heat deflection (HDT **250 °C** dry), it absorbs **5.2–5.8% water at saturation**. lookpolymers In a salon where tools are routinely immersed in Barbicide solution and exposed to water throughout the day, this moisture uptake causes **dimensional swelling that will loosen connector tolerances over time**. Published research shows fatigue strength drops **30–50%** from dry to conditioned state in PA66-GF30, and glass-fiber-matrix debonding at fiber ends is the primary failure mechanism. For a connector interface designed to ±0.05 mm tolerances with 0.10–0.15 mm clearance, moisture-induced swelling is a reliability risk.

Property	POM Copolymer	PA66-GF30	PEI (Ultem 1000)	PPSU (Radel)
Cost ($/kg)	**$2.10–3.50**	$4.00–8.00	$100–250	$30–80
Water absorption (saturation)	**0.22%**	5.2–5.8%	0.25%	0.37%
Flexural modulus (MPa)	2,585	**6,500**	3,447	2,410
HDT at 1.8 MPa (°C)	136	**250**	204	207
Barbicide resistance	Excellent	Good	Fair (ESC >1 hr)	Excellent
Bleach/H_2O_2 resistance	Moderate	Moderate	Good	Good
Fatigue endurance (10^7 cycles)	**23–30 MPa**	Moisture-sensitive	—	—
Snap-fit max repeated strain	**3.6%**	0.9–1.2%	—	—
Cost per 40 g unit (material)	**$0.08–0.14**	$0.16–0.32	$4.00–10.00	$1.20–3.20

Recommendation: Use POM homopolymer (Delrin) for handles as well as comb bodies, or consider **glass-filled POM (POM-GF25)** for the handles if additional stiffness is needed. POM homopolymer has fatigue endurance of **30 MPa at 10^7 cycles** Midlandplastics and superior wear properties for the connector rail surfaces (coefficient of friction ~0.2, self-lubricating). AZoM If the handles require stiffness beyond what POM provides, a POM-GF composite or short-carbon-fiber-filled POM would deliver increased modulus without the moisture liability of PA66.

Y.S. Park's use of PEI/Ultem Royal Barber Supply GHD ($100–250/kg) carries a material cost of **$4–10 per comb** versus **$0.08–0.14 per comb** in POM. PEI's value lies in its 220 °C glass transition temperature laminatedplastics and 80% elongation at break, laminatedplastics which gives thin teeth extreme flexibility. However, PEI exhibits **environmental stress cracking in Barbicide** with exposure beyond one hour — Y.S. Park explicitly warns against prolonged immersion. YS Park +2 POM is the safer choice for a tool that will be repeatedly disinfected.

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4. Ergonomics validate weight but flag length and fork width

The system's **22–35 g weight target** is ergonomically excellent. Published guidelines (CCOHS, NIOSH, EHS Today) set the upper limit for precision tools at **400 g**, CCOHS making TAYLKOMB's target roughly **1/12th** of the threshold. Professional cutting combs weigh 9–30 g across the benchmarked brands, so this range is consistent with market expectations.

However, the **405–424 mm assembled length is a serious ergonomic outlier**. Professional cutting combs top out at **222 mm** (Sam Villa Long Cutting). Even the longest tail comb found — Y.S. Park 106 — measures only **245 mm**. YS Park +2 Flat irons, the longest common hand-held styling tools, run **229–318 mm**. TAYLKOMB's assembled length exceeds even flat irons by 100+ mm.

Published studies on hairdresser musculoskeletal disorders are alarming: **hand/wrist MSD 12-month prevalence ranges from 11% to 53%**, PubMed Central BioMed Central and hairdressers suffer carpal tunnel syndrome at **5× the rate** of the general population Cosmetology Guru (Kozak et al. 2019, *Journal of Occupational Medicine and Toxicology*; Saini et al. 2022). Round-brush styling generates **50 repetitions per minute** for the dominant hand — exceeding Kilbom's high-risk threshold of 10 rep/min. PubMed Central Longer tools create larger moment arms that amplify wrist torque during these angular movements, making excess length a genuine occupational health concern.

The **fork/double handle at 27.2 mm outer width** is marginal. Cal/OSHA specifies a **minimum of 25.4 mm (1 inch)** for double-handle precision tools. CA The TAYLKOMB design clears this by only 1.8 mm, and the internal grip width between prongs is narrower still. Published pinch-grip research shows comfort and strength peak at **45–55 mm** grip span (Pheasant & Haslegrave). CCS Construction **Widening to 30–35 mm outer minimum** is recommended.

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5. The connector interface is well-sized but needs material revision

The **61 × 45.2 × 20.32 mm** connector interface provides an engagement length-to-height ratio of **3.0×** (61/20.32) — well above the 1.5× minimum for wobble prevention. The engagement length-to-width ratio is **1.35×** (61/45.2), borderline but adequate when combined with a ball-detent providing a preloaded registration point at the end of travel.

For the dovetail rail, a **60° included angle** is recommended — it is the industry standard for smaller profiles, provides equal play in horizontal and vertical directions, and is compatible with standard injection-mold tooling. Critical tolerances should be held to **±0.05 mm** (achievable with precision injection molding Formlabs Protolabs Network per ISO 20457:2018), with designed clearance of **0.10–0.15 mm per side** for smooth sliding without wobble.

Ball-detent specifications for the locking mechanism should target **3–5 mm stainless steel ball diameter** with **5–15 N engagement force** for comfortable hand operation. A 90° conical detent notch in POM will provide a satisfying click. Using a commercial press-fit stainless steel ball plunger in the handle engaging a molded POM detent pocket achieves **10,000+ cycle life** with minimal wear, because POM's low friction coefficient (0.15–0.25) and self-lubricating properties make it the industry-standard detent-pocket material.

POM snap-fit elements can exceed 100,000 cycles at strains below 3%, with a published fatigue endurance limit of 30 MPa at 10^7 cycles. PA66-GF30 snap-fits, by contrast, must stay below **1% strain** for 10,000+ cycles — a much tighter design window exacerbated by moisture-induced degradation. This further supports switching the handle material to POM or a POM-based compound.

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6. Injection molding confirms all critical features are manufacturable

Comb tooth thickness of 1.0 mm at the tip is feasible in POM. DuPont's Delrin design guide lists a minimum wall thickness of **0.75 mm**, and Y.S. Park's PEI combs demonstrate ~1 mm tooth tips in commercial production. Tooth roots should taper to **1.5–2.0 mm** for structural integrity. PA66-GF30 is not recommended for comb teeth — glass fibers create surface roughness and reduce impact strength in sections below 1.0 mm.

The **1.3 mm rat-tail tip diameter is feasible** and sits comfortably above the 0.75 mm minimum feature size for POM injection molding. Adequate draft (1–2°) and proper venting at the tip in the mold tool will ensure complete fill. Existing professional rat-tail combs achieve tips of **1.0–2.0 mm**, placing this design squarely within the proven production range.

Structural comb spines should be **2.0–2.5 mm wall thickness** in POM, with optional ribs at **40–60% of wall thickness** (0.8–1.2 mm) for added stiffness without sink marks. Handle spines in POM can be designed at 2.0–2.5 mm as well — if the handles were in PA66-GF30, the higher modulus would have permitted 1.5–2.0 mm, but the material switch to POM means slightly thicker walls are needed for equivalent rigidity.

Production tolerances for the rail/slot interface are achievable at **±0.05 mm** using precision injection molding Formlabs with conformal cooling. POM's shrinkage (**1.8–2.2%**) is significant but highly predictable and consistent, AZoM enabling accurate mold compensation. At production volumes of 5,000 units, estimated per-unit cost for POM parts is **$4–8** including tooling amortization, material, and processing — roughly half the cost of an equivalent PEI part.

Validation verdict: TAYLKOMB specs vs. market and engineering benchmarks

Specification	TAYLKOMB Current	Benchmark / Standard	Verdict
Main Comb body length	208 mm	Cutting combs: 181–222 mm	✅ **Within envelope**
Comb overall (with connector)	229 mm	Tail combs with handle: 203–245 mm	✅ **Within envelope**
Assembled length (comb + handle)	405–424 mm	Longest comb: 245 mm; flat irons: 229–318 mm	❌ **2× too long — redesign connector or shorten handles**
Round Handle overall	243 mm	Handle standards: 100–152 mm recommended	⚠️ **Long but acceptable if most length is working section**
		Production range:	

Rat-tail tip diameter	1.3 mm	1.0–2.0 mm; POM minimum: 0.75 mm	✅ **Feasible**
Flat Handle overall	256 mm	Handle standards: 100–152 mm	⚠️ **Long — assess if excess length adds function**
Fork Handle outer width	27.2 mm	Cal/OSHA minimum: 25.4 mm; comfort optimum: 45–55 mm	⚠️ **Marginal — widen to ≥30 mm**
Fork prong cross-section	5.43 × 5.6 mm	Precision grip: 8–16 mm diameter recommended	⚠️ **Thin — consider widening for sustained grip comfort**
Connector interface	61 × 45.2 × 20.32 mm	Engagement ratio ≥1.5× cross-section	✅ **Adequate (3× height, 1.35× width)**
Weight target	22–35 g assembled	Combs: 9–30 g; precision tool limit: 400 g	✅ **Excellent**
POM for comb bodies	✓	Used by Krest (Delrin), Pearlon Hercules Sägemann	✅ **Industry-validated material**
PA66-GF30 for handles	✓	Water absorption 5.2–5.8% at saturation	❌ **Reject — moisture swelling will loosen connectors; switch to POM or POM-GF**
Tooth thickness (implied 1 mm tips)	~1 mm	POM minimum: 0.75 mm; Y.S. Park achieves ~1 mm	✅ **Feasible**
Spine wall thickness	Not specified	POM recommended: 2.0–2.5 mm	ℹ️ **Target 2.0–2.5 mm**
Rail tolerance	Not specified	±0.05 mm precision molding, 0.10–0.15 mm clearance/side	ℹ️ **Specify ±0.05 mm on critical surfaces**
Heat resistance (POM)	82–100 °C continuous	Y.S. Park PEI: 220 °C; Sam Villa: 232 °C Amazon	⚠️ **POM is lower — market combs positioned as heat-resistant; consider this in positioning**

Critical action items for CAD redesign

1. Reduce assembled length to ≤300 mm. The 405–424 mm range has no precedent in professional combs and exceeds even flat irons. Options: shorten the connector interface depth, reduce handle working sections, or introduce a telescoping/nesting feature. A target of **250–280 mm assembled** would bring the tool into the long-tail-comb range while preserving modularity.

2. Replace PA66-GF30 handles with POM homopolymer or POM-GF25. The salon environment's daily water and chemical exposure makes hygroscopic PA66 dimensionally unreliable at the tolerances the connector requires. POM homopolymer (Delrin 500P) provides superior fatigue endurance (30 MPa at 10^7 cycles), Midlandplastics self-lubricating wear surfaces, and only 0.25% water absorption. Midlandplastics If additional stiffness is needed, glass-filled POM or carbon-fiber-filled POM eliminates the moisture risk entirely while maintaining adequate modulus.

3. Widen the fork handle outer width to ≥30 mm, ideally 32–35 mm. The current 27.2 mm barely clears the ergonomic minimum and leaves insufficient inner grip width for sustained professional use.

4. Specify connector tolerances explicitly in CAD. Target **±0.05 mm** on dovetail rail mating surfaces with **0.125 mm nominal clearance per side**. Use a **60° dovetail angle** and specify a **3–5 mm stainless steel ball plunger** at 8–12 N engagement force with a 90° conical POM detent pocket.

5. Address heat-resistance positioning. POM's continuous service temperature of 82–100 °C is significantly below MakerStage the 220–232 °C ratings of Y.S. Park YS Park and Sam Villa. Amazon YS Park If TAYLKOMB combs will contact hot irons during styling, either add a heat-resistance disclaimer or consider PEI or PPSU for the comb bodies at the premium tier. PPSU ($30–80/kg) offers a middle ground: **207 °C HDT, 220 °C Tg, excellent Barbicide resistance**, and roughly one-third the cost of PEI.

Conclusion

TAYLKOMB occupies a genuinely uncontested market position — no modular comb system exists anywhere in the professional beauty industry. The core engineering concept is sound: a rail-and-detent interface in POM can achieve **10,000+ connection cycles** with tight tolerances, and the 61 mm engagement length provides adequate wobble resistance. The weight target and comb-body dimensions align precisely with the competitive envelope established by Y.S. Park, Sam Villa, and Hercules Sägemann.

The two issues that must be resolved before tooling are assembly length and handle material. A 405 mm comb is not a comb any stylist has ever held; bringing it under 300 mm is essential for market acceptance and ergonomic safety. PA66-GF30's moisture absorption lookpolymers in a wet salon environment will gradually degrade the precision connector fit that makes the modular concept work — POM eliminates this failure mode entirely. With these corrections, the system's specs are well-supported by engineering data and competitive benchmarks for CAD finalization and prototyping.